AI Usage Journal

AI Tools Used (as a team)

NotebookLM

NotebookLM was used to organise and summarise sources so as to save time. It was also great for generating study guides about individual sources/groups of sources, and eye-catching infographics to help understand core concepts.

Claude Sonnet 4.6

This was used by the team for multiple tasks related to our understanding of the topics. Using Claude helped with brainstorming ideas, confirming concepts, and overall guiding our thought process. It was also used for structuring the notes and the quiz.

GitHub Copilot Chat (Auto: GPT-5.4 mini, GPT-5 mini)

Copilot was primarily used to debug python code for the annotated code walkthrough. Additionally, it also assisted in building the simulator by generating code specific to the framework, and explained further about how Streamlit functions.

Sarah Demanuele

Section 2:

Prompt: “Suggest analogies to explain the need for gamma correction”

Summary: Several analogies were suggested including speaker volume and staircase spacing. As a team we chose the speaker volume analogy because it explained logarithmic perception intuitively without losing technicality. Before this prompt we had planned to open with the power law equation directly and this changed that decision entirely. The analogy now appears in the study notes under "Analogy Comparison" to introduce the mismatch between linear light and human perception before any technical content appeared.

Prompt: “What is the key conceptual difference between gamma correction and tone mapping?”

Summary: The response explained that gamma correction solves a perception problem, aligning linear data with how humans see, while tone mapping solves a range problem, compressing HDR into LDR. This was a turning point because our draft had described tone mapping as an extension of gamma correction, which was wrong. The distinction is now clearly understood and reflected in our work.

Prompt: “Can you explain the difference between encoding and decoding in gamma correction specifically why do they need eachother?”

Summary: Initially, we understood encoding and decoding to be unrelated processes (i.e. either one or the other is used). However, the response clarified that they are a matched pair. This interaction changes our understanding as we initially treated encoding and decoding as independent processes rather than as a pair.

Prompt: “I have uploaded our completed study notes. Generate a mindmap of all the key concepts covered so I can verify nothing important has been missed before moving on to the other deliverables”

Summary: NotebookLM generated a mindmap from the completed study notes, extracting and organising the key concepts. This was originally intended to confirm all key concepts had been covered in the notes. However the mindmap output was detailed enough to use directly as a structural blueprint, for the cheat sheet and presentation slides. What started as a checking step became the most practically useful AI output in the project.

Section 3:

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

The hardest concept to explain was the motivation behind gamma correction. Different sources gave different explanations. Some sources explain that it was developed because humans perceive light logarithmically, while others said it exists because CRT monitors respond non-linearly to voltage. In the end we chose to present gamma correction as a fix for human perception, while introducing CRT history as context.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

An instance where AI provided something we thought was complete but was regarding the main motivation of gamma correction. Initially, the LLM was not mentioning CRT monitors and how gamma correction was developed initially for that use. This led us to research more and find more academic sources to both explain and support the motivation.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

One moment that comes to mind is when researching for alternatives of gamma correction and tone mapping. Initially, we searched for alternatives on google but all the search results included alternatives for a presentation maker called GammaAI. Therefore we used Claude, which hallucinated, in fact saying that Reinhardt and ACES were alternatives for Tone Mapping, and that Tone Mapping was an alternative for Gamma Correction. After further prompting, we got some results we were looking for. This proved the importance of cross-referencing AI outputs with independent research before drawing conclusions.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

The most useful moment was using NotebookLM to summarise and extract key points from research papers and our own study notes. I then uploaded the study notes we created to NotebookLM to check that we mentioned all key concepts, through the generation of a mindmap. This mindmap was also used as the blueprint for the reference cheat sheet and for our own presentation slides.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Even after completing the Learning Pack, we still would not feel fully confident explaining the exact mathematical and perceptual differences between tone mapping operators such as Reinhard, ACES, and Hable at an advanced level. This is probably due to us not going into much detail about such topics in the notes.

Reuben Mifsud

SECTION 2 - NOTABLE PROMPTS:

Prompt: “You are a bachelors student studying AI and these are your notes for a new topic of ‘computer vision for AI’. How well would you understand this (having no prior knowledge about it)? Which areas need reinforcement, which are perfect as they are? Give me your full feedback”
Summary: The AI started by giving us a general ranking of the notes (from “needs work” to “near-perfect”) and then followed on by specifying certain areas which could be improved. This helped us get a better understanding on where our notes were strong and easy to understand, while knowing how to improve the notes. Since I had set my chatbot to not sugarcoat things it was direct and straightforward with its answers, letting me know when I did wrong yet still showing me approval for my rightdoings.

Prompt: “Explain gamma correction to me as if I have no background in image processing whatsoever. What is its purpose? How does it work? Do not include complicated technical terms”
Summary: This prompt was asked in 2 different conversations - My main one for this project and a fresh conversation (as mentioned in question 2 of Section 3). The reason for this is to detect any discrepancies between the two conversations, so to confirm whether the AI is answering truthfully or hallucinating. The AI was in fact hallucinating as it said CRT monitors were the reason for gamma correction on one conversation, whilst in the other conversation it mentioned human perception. This led me to research more on the matter so as to get as close to the truth as possible.

Prompt: “Aren’t ACES and Reinhard simply types of tone mapping operators? I don’t believe they are alternatives to tone mapping as you said… Please clarify this.”
Summary: After responses like these, the AI would either attempt to prove itself right, or it would agree with me and try to clarify the issue but fail to do so. In return, I would always end up more confused than I was before. In such cases, I sometimes tackled the issue by opening a new chat specifically about the issue, or by researching more online. In this specific case, online websites and sources about alternatives for tone mapping operators seemed misleading and so I chose to rely more on AI that time.

Prompt: "My notes currently state that gamma correction exists to compensate for human logarithmic perception. However, after research I believe this is inaccurate because gamma correction originated because of CRT monitor nonlinearity, not human vision. Can you confirm this and help me restructure the notes accordingly?”
Summary: The intent of this prompt was to challenge the AI and compare its response with the research I had done outside of the conversation. The AI confirmed my correction and agreed that the CRT origin story was the historically accurate primary motivation, with human perception being the coincidental side effect. This led me to restructure the notes so that the true reason for gamma correction’s existence was evident.

SECTION 3 - FIVE QUESTIONS:

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

Personally, I found gamma correction to be the hardest concept to explain, not because the concept itself was difficult to explain, but rather because the easiest way to explain it was by presenting it as a solution to the nonlinear human perception of light, when in reality the reason behind gamma correction was more complicated (CRT brightness proportional to voltage). By consulting with AI, I reached the conclusion to still present gamma correction as a fix to the human perception of light, which is true, and later on explaining the more detailed actual historical reason as to why gamma correction exists.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

This only happened once for me and it was when I asked for clarification on why gamma correction exists. I opted to ask this in 2 separate Claude conversations - the main one which I was using for this project, and a fresh conversation. Their responses to the same question were polar opposites, which led me to dive deeper in researching papers to really nail my understanding of the concept.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

This would be the same moment as mentioned in question 2, where I got two completely different answers from the same LLM which were given the same question. This taught me to always fact-check answers made by AI, especially if the answer is vague. Moreover, the best thing to do when in a deep hole talking to and fro with AI is to simply take a break and forget about the topic, then start fresh again.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

I think the most notable moment where AI made our team faster was when used to search for research papers. Using NotebookLM for this made extracting information from sources infinitely faster and easier for us, and our team could simply read about the main sections of the paper to get a clear idea of it, without having to read the paper entirely.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Building a multi-faceted learning pack including notes, a quiz, a code walkthrough, and a simulator, naturally demands deep understanding of the topic. I stand by what Feynman said, which is: "write to understand, teach to master." Having done both, I believe our grasp of the topic is strong. With that being said, sub-topics outside the scope of our pack (such as the internal mechanics of CRT displays or the historical development of tone mapping operators) would most probably not be answered with full confidence, because our team had not focused specifically on these for the learning pack.

Nicholai Camilleri

Claude Sonnet 4.6:

- Asked for a start to finding sources, general understanding about the topic and code help.

NotebookLM:

- Used by the entire team in order to understand and query sources.

Copilot Built into VSCode (Model is automatically assigned by the program):

- Very little use, just to help with a few syntactical issues.

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

    Tone Mapping was the hardest part to explain as you can't fully visualise it. In trying to present an original HDR image and a tone mapped one, both need to be tone mapped to be displayed on a screen regardless. This presented an issue as given this topic is about images, visuals are the best way to communicate changes.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?

    Generally AI either adds bugs to code or adds unnecessary code that complicates the program more, making it more prone to bugs. Situations when this happened are for sure when I asked for help regarding the tone mapping operators. AI would give fully fleshed out code, using functions and adds multiple options whilst a simple google search would show you that an equation you do in one line has the same effect if not a better one on an image.

    Prompt: Are you absolutely sure this is correct, give me your sources

    Context: I was attempting to compare HDR and LDR in a histogram of range 0 to 255, however this clearly wouldn't make much sense to explain as the entire hdr image is all between the 0 to 1 bin whilst the LDR image was distributed as expected. I was asking AI to verify the explanation to me, which I checked the sources it gave me and searched separately to check.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

Prompt: Could you replace this with our own power_law function above? Make it do the same thing

Summary: I was asking AI to change code it had given me regarding the power law to make it a built function rather than using a built in function in a library, this was specifically to make a power law function that takes HDR images or images with RGB values already normalised to the 0 to 1 range.

This example was a waste of time as the AI barely understood me and tackling the problem myself by adapting my previous power law function took a couple minutes and produced a better result. Situations like this do show me that trying to tackle the problem myself is generally better, as the AI requires so much context explained in advance to get the result that you want that generally, when it is related to small issues or small tasks, it is not worth the time. That being said, I could have prompted the AI better to give me a more tailored solution.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

Personally, AI helped best when finding sources and compiling them. Asking AI for papers tailored to your needs helps greatly and speeds up the research process greatly. On top of that, AI like NotebookLM further make understanding these papers easier and any misunderstandings you have can be queried directly against these papers thanks to NotebookLM.

Prompt: (Point Processing - Gamma correction and tone mapping
Power law, perceptual motivation, non-linear encoding, HDR tone mapping at a conceptual level.)

Can you individually describe to me the above because I am quite uninformed about these and would like a starting point to eventually start going more in depth. Moreover, suggest other sources and papers to help me

Context: This was at the very beginning of the assignment were I was understanding what we had to teach. Personally having this very brief explanation to read through before going through more in depth sources helps me be mentally prepared and helps me have an idea of what to expect.

Prompt: How would I know whether a source is a valid and adds to my assignment or not? For example, this one does implement tone mapping for HDR and discusses how the eye adapts to light, it is 2005 paper.

https://jamesferwerda.com/wp-content/uploads/2015/06/r06_irawan05_egsr.pdf

However, my main issue is that do I take this paper as common practice of tone mapping or do I just assume it is someone's implementation to a problem? How do I approach this

Context: Still in the beginning of the assignment, I wanted a few more guidelines on how to search and validate sources. As I asked in the prompt, I was unsure whether to take knowledge from some papers and assume them as common practice given how many experimental papers are released in this field. AI here really helped and explained multiple ways and things to look out for in order to help me determine what papers are worth having a look at.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

My understanding in Tone Mapping has gotten better however, questions relating to it are probably my weak points. Having coded a chunk of the tone mapping code myself I understand what goes on, but to effectively compare the default image to the converted one and explain what and how change still challenges me.

Alexander Busuttil Cazac

AI Tools Used

I personally made use of NotebookLM and Copilot throughout the project. You may find descriptions of their use at the beginning of this journal.

Notable Prompts

A. Tool: NotebookLM, with a source selected.

   a. Prompt: "*What does this touch up on? Is it relevant for our gamma correction and tone mapping project?*"

   b. Summarised Response: NotebookLM proceeded to highlight the source's explanations and studies on gamma correction and tone mapping operators.

B. Tool: Copilot, with the scalar constant C selected in the power law code.

   a. Prompt: "*Add this option to gamma correction as a slider*"

   b. Summarised Response: Copilot proceeded to create a new slider with range 1 - 255, and also suggested unnecessary changes to unrelated portions of the simulator's code.

C. Tool: Copilot, with the annotated walkthrough notebook and simulator file attached.

   a. Prompt: "Take a look at these two. Change requirements.txt to reflect all needed packages"

   b. Summarised Response: Copilot suggested changes to requirements.txt, updating it with (most of) the required packages for the python code around the repository.

D. Tool: Copilot, with the simulator file attached.

   a. Prompt: (Pasted Exception)

```
"File
"/home/alec/Documents/GitHub/ARI2129-Computer-Vision-Group-Proje
ct/Simulator.py", line 181, in <module>

main()

...
```

```
File
"/home/alec/.conda/envs/CVT/lib/python3.12/site-packages/streaml
it/elements/widgets/slider.py", line 942, in _slider

raise StreamlitAPIException(msg)"
```

(shortened)

b. Summarised Response: Copilot suggests changes to Streamlit-specific code: "That traceback is a StreamlitAPIException raised when a widget (here st.slider) received an invalid value."

Five Questions

1. *Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?*

   Explaining that ALL images should be tone mapped before being displayed onto an LDR display was not easy. The difficulty primarily came from a novel yet incomplete understanding of HDR technology. Additionally, tone mapping operators and HDR technology in general present new concepts which have not been covered thoroughly in university material, whereas gamma correction and the power law are more trivial concepts which were also explained through the tutorial notebooks.

2. *Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?*

   While Copilot was correct most of the time, it did have a few minor hiccups. The best example of this was when I asked it to read the annotated walkthrough notebook and simulator code, and asked it to print all required packages into the requirements.txt file (prompt C). It turned out that it forgot the ipykernel package, but I did not spot this mistake.

   In order to validate this particular response, I asked Nicholai Camilleri to separately clone the repository into a new location, run the command to install ONLY the packages in requirements.txt, and try running both the notebook and the simulator. This was where the mistake was spotted.

3. *Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?*

   An example of time-wasting was when I asked Copilot to add a slider for the constant scale 'c' in the power law function (prompt B). While it did successfully add a slider for the argument, it also edited completely unrelated sections of the code, which were completely unwanted and unnecessary. This issue kept happening consistently, until I finally asked it to stop suggesting the same unwanted change.

I realised through this chain of instances, as well as others outside of this project, that having better knowledge of prompt engineering techniques is essential for having an AI assistant work alongside, but also that sometimes a simple change may be easier than one thinks, and therefore may not require prompting an LLM.

4. *Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?*

   A good example of increased efficiency using AI was research (prompt A). NotebookLM likely saved multiple hours of time by summarising sources and making them more digestible, all while linking evidence (citing) to specific portions of the respective source material. This was also true for debugging the simulator's exceptions, where not only was Copilot vital in fixing problems (prompt D), it was also a good teacher. It helped me learn Streamlit, which I was completely new to.

   Copilot also generated certain sections of the simulator code (Streamlit). This was very useful, as it wrote a reasonable amount of boiler-plate code for me, with far better quality than I would have written manually.

5. *What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?*

   I think that while we did a good job in covering our self-assigned sub-topic, we may not have covered the concept of non-linear encoding enough. Therefore, a simple yet vague question such as "Why does non-linear encoding of images in a machine matter" may puzzle us.

Zack Haber

Section 1: Tools Declared

Claude Sonnet 4.6:

- Improve sentence language in some of the definitions so that the notes are easier to understand for new learners.
- Used to get the get the definitions and brief understanding (Not detailed) of:
  - Reinhard
  - ACES Filmic
  - Hables
- Helped in generating a few of the questions for the quiz (Not all, only around 2) whilst also providing the options, answer and reasons as to why the correct answer is correct and how the other options could be interpreted as the correct answer, potentially confusing the reader and making them guess incorrectly.

NotebookLM:

- Helped in organising and finding useful research papers based on the topics which we required.
- Helped in summarising the papers into brief paragraphs, this helped us in:
  - Understanding properly the aim of each research paper
  - Helped us come up with useful definitions for topics within the notes, and obtained useful data such as equations and graphs.

Section 2: Notable Prompts

1. **Prompt**: "I have attached a quiz I am working on, I currently have 8 questions complete and need 10 in total, do you have any suggestions?"

   Brief:
   I had asked AI to suggest 2 additional quiz questions to complete a 10-question set on tone mapping, matching the existing format with multiple choice answers, correct answer labels, and feedback for correct and incorrect options. It had generated the last two questions:

- “A typical tone mapping operator applies a non-linear curve that is:”
- “What is a luminance?”

  For each of the questions I was given a few options (Some of these were changed to my own liking) whilst also providing why some of the questions could be misleading and what they actually mean.

2. **Prompt**: “What are the What, Why and Hows of Tone Mapping? Explain in simple terms”

Answer: Here's a structured breakdown of the **What, Why, and How** of Tone Mapping:

What is Tone Mapping?

Tone mapping is a technique used in image and video processing to translate the wide luminance range of High Dynamic Range (HDR) content into a range that can be accurately displayed on Standard Dynamic Range (SDR) or lower-capability HDR screens. It operates on the brightness values of individual pixels, curves, or regions of an image, reshaping the tonal distribution so that both shadow detail and highlight detail survive the conversion. It exists in two main forms: static tone mapping, which applies a fixed curve based on content metadata, and dynamic tone mapping, which continuously adjusts its parameters in real time to suit each scene or frame.

Why is Tone Mapping Needed?

The core problem tone mapping solves is the enormous gap between how content is mastered and how most displays can actually reproduce it. Modern HDR content is often graded at peak brightness levels of 1,000 to 4,000 nits, while the vast majority of consumer TVs and monitors are only capable of outputting 100 to 500 nits. Without tone mapping, this mismatch causes highlights to clip — losing all detail and appearing as flat, blown-out white — and shadows to crush into solid black, stripping away texture and depth. Beyond technical correction, tone mapping also serves a creative purpose: it preserves the director's or colorist's intent by maintaining the relative contrast and mood of a scene even when the absolute brightness values must be reduced.

How Does Tone Mapping Work?

At its core, tone mapping applies a transfer function — commonly called a tone curve — to the luminance channel of an image. Brighter values are compressed more aggressively than mid-tones, and shadows are gently lifted or preserved, creating a natural-looking roll-off rather than a hard cut-off. More advanced implementations separate the image into local regions and apply different curves to each area, allowing a bright sky and a dark foreground to each retain detail simultaneously. Dynamic tone mapping goes further still, analyzing the histogram of each frame and adjusting the curve on the fly, ensuring that a sudden cut from a dark interior to a bright outdoor scene is handled smoothly without either end of the tonal range being sacrificed."

Section 3: Five Questions

1. Which concept in your topic did you find hardest to explain to someone else, and why do you think that was?

I was having a hard time properly explaining Tone Mapping since it's hard to properly visualise the difference between an HDR image with tone mapping and one without. This was due to the fact that both of the images have to be tone mapped to be able to see the images themselves.

I had also found Gamma Correction a bit tricky as I was having a bit of a hard time actually understanding the topic itself.

2. Where did AI give you something that looked correct but turned out to be wrong or incomplete? What did you do about it?
We had first used AI to get an understanding on Gamma Correction. After some time, we noticed that we had been given a misconception about the topic. We had assumed it was because humans see logarithmically, but in reality, it exists because some display screens work non-linearly, which was the true reason why gamma correction exists, it just so happens to work with human perceived light. So the AI was somewhat correct but not 100% correct in this aspect.

3. Describe a specific moment where using AI wasted your time. What happened and what did you learn from it?

When I was coming up with the questions for the quiz, I kept the AI for ideas for some of the questions. But for some reason, the AI just kept giving me questions, answers and definitions which I had already previously given him. Furthermore, the new questions which the AI had given which I never had were too complicated and not even I could understand them properly.

4. Describe a specific moment where AI genuinely made your team faster or produced something better than you would have managed alone. What made the difference?

The most useful moment from AI was definitely when it was used to find useful research papers which we were able to use for our notes and gain extra knowledge. We then proceeded to use AI to get a general idea of what the research papers are about and to help highlight any key information which would really help with our project.

Getting a brief summary about each of these papers helped us better understand the topic whilst also making note writing much easier for us all.

5. What question about your topic does your team still not feel confident answering, even after building the entire Learning Pack?

Our team has grasped a lot from creating this learning pack. However due to having a large group, we naturally split up to work on other components separately to build this project quicker. So whilst I don't think there is one big question that none of us can answer, if we were to receive basic questions based on other areas in the project that we did not work on, we would have a hard time finding the right answers for them (Ex: I did not take part in constructing the simulator, any questions related would be a challenge for me to answer).